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Atmospheric Thermodynamics

Chapter 3: The Second law of Thermodynamics

Objective

- *To Know the statement of Second law of Thermodynamics*
- *To Explain The Carnot cycle*
- *To Know Carnot Efficiency and Its Connection to Meteorology*

First Law of Thermodynamics (Recall)

- The first law of thermodynamics is based on energy conservation. It states that energy cannot be created or destroyed, only transformed (e.g., heat to work, work to heat).
- It does NOT tell us the direction of processes.
- It does not restrict how energy is transformed. It would allow processes that never happen in reality

Heat Flow in the Atmosphere

What we observe:

- Heat flows from warm surface → cooler air
- Heat flows from tropics → poles

What first law allows:

Heat could also flow from cold air → warm surface

But this never happens naturally

Second law says:

Heat flows spontaneously from warm to cold, not the reverse.

The Second law of Thermodynamics

- The second law of thermodynamics, is concerned with the maximum fraction of a quantity of heat that can be converted into work.
- This conversion was first clearly demonstrated by Carnot, Who also introduced the important concepts of cyclic and reversible processes.

Statement Of The Second Law Of Thermodynamics

- Heat naturally flows from a hotter body to a colder body and never spontaneously in the reverse direction.
- The total entropy of an isolated system always increases over time (entropy represents disorder).
- No process can be 100% efficient in converting heat energy into work

Meteorological Interpretation

- Heat transfer in the atmosphere follows this principle, such as warm air rising and cold air sinking (convection).
- In any natural process, the total entropy of an isolated system will either increase or remain constant, but it will never decrease, which influences how energy is distributed.
- The efficiency of energy transfer in weather systems, such as tropical cyclones, is limited by entropy

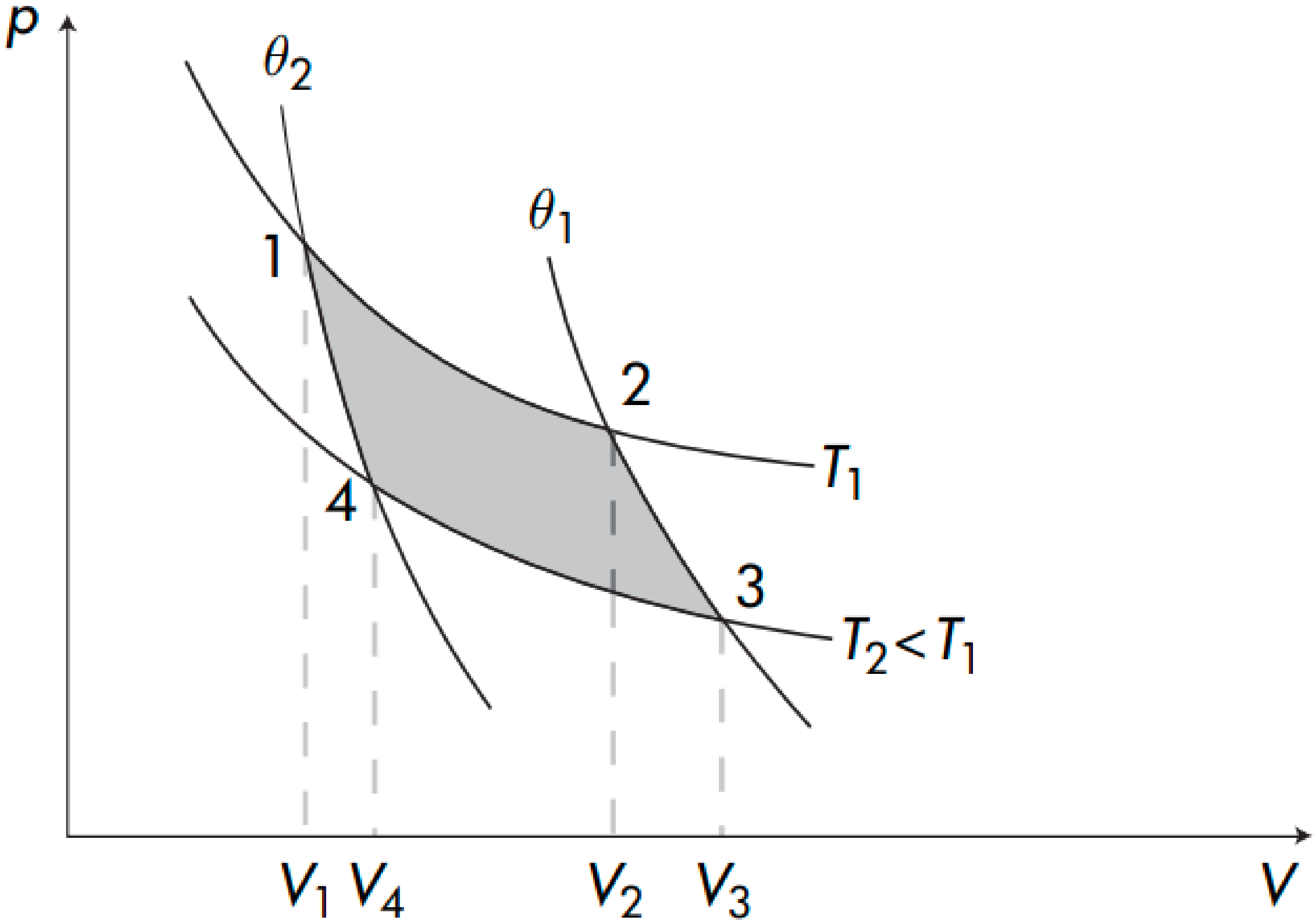
Cont. . .

- The Second Law of Thermodynamics is essential in meteorology because it governs energy transfer, atmospheric circulation, and cloud formation.
- It defines the direction of heat flow, entropy changes, and efficiency limits of atmospheric processes.

The Carnot Cycle

- The Carnot cycle is a thermal engine that receives heat from some source and transforms into work.
- In thermodynamical terms the Carnot cycle is a cyclic transformation consisting of the following four steps (See figure below)

Cont. . .



Steps in a Carnot cycle

Cont. . .

- ❖Step 1: reversible isothermal expansion $(1 \rightarrow 2), T = T_1 = Constant$
- ❖Step 2: reversible adiabatic expansion $(2 \rightarrow 3), \theta = \theta_1 = Constant$
- ❖Step 3: reversible isothermal compression $(3 \rightarrow 4), T = T_2 = Constant$
- ❖Step 4: reversible adiabatic compression $(4 \rightarrow 1), \theta = \theta_2 = Constant$
  - ✓with $T_2 < T_1, \theta_2 < \theta_1$.

Cont. . .

- Let us now calculate the work done and heat absorbed when an ideal gas is subjected to this cycle.
- The change in internal energy is zero because the overall transformation is a cyclic one.
- Let’s consider each step separately

Cont. . .

❖**Step 1**

$\Delta T = 0$, because it is isothermal process

$$W_{12} = \int_{V_1}^{V_2} pdV = nR^*T_1 \int_{V_1}^{V_2} \frac{dV}{V} = nR^*T_1 \ln \frac{V_2}{V_1}$$

$$\Delta U_{12} = C_v \Delta T = 0$$

$$Q_{12} = W_{12}$$

➢Since $V_2 > V_1$ it follows that $W_{12}, Q_{12} > 0$ indicating that during step 1 work was done by the gas and heat was absorbed by the gas from the heat source at temperature T_1.

Cont. . .

❖**Step 2**

$$Q_{23} = 0$$

$$\Delta U_{23} = C_v(T_2 - T_1) < 0$$

$$W_{23} = -\Delta U_{23} = -C_v(T_2 - T_1) > 0$$

- Therefore work done is positive, which implies that it is expansion.

❖**Step 3**

$$\Delta T = 0$$

$$W_{34} = \int_{V_3}^{V_4} pdV = nR^* T_2 \ln \frac{V_4}{V_3}$$

$$\Delta U_{34} = 0$$

$$Q_{34} = W_{34}$$

- Since $V_4 < V_3$ it follows that $W_{34}, Q_{34} < 0$ indicating that now work is done on the gas and heat is given away by the gas to the source at temperature T_2

Cont. . .

❖ **Step 4**

$$Q_{41} = 0$$

$$\Delta U_{41} = C_v(T_1 - T_2) > 0$$

$$W_{41} = -\Delta U_{41} = -C_v(T_1 - T_2) < 0$$

- Therefore work done is negative, which implies that it is compression.

- From the above it follows that the total work done is

$$W = W_{12} + W_{23} + W_{34} + W_{41}$$

$$= nR^{*}T_1 \ln\frac{V_2}{V_1} - nR^{*}T_2 \ln\frac{V_3}{V_4} \quad \dots\dots\dots\dots (1)$$

Cont. . .

- Because transformations $2 \to 3$ and $4 \to 1$ are adiabatic the following relations apply:

$$T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1} \quad \text{and}$$

$$T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1}$$

- By dividing the above two equations we have

$$\frac{V_2}{V_1} = \frac{V_3}{V_4}$$

Cont. . .

❖Accordingly, equation (1) can be written as

$$W = nR^*T_1 \ln\frac{V_2}{V_1} - nR^*T_2 \ln\frac{V_2}{V_1}$$

$$W = nR^*\left(\ln\frac{V_2}{V_1}\right)(T_1 - T_2) \dots\dots\dots\dots (2)$$

❖For simplicity we will now denote the amount of heat absorbed by the gas from the heat source at T_1 (i.e. Q_{12}) as Q_1 and the amount of heat given by the gas to the source at T_2 (i.e. Q_{34}) as Q_2 .

❖Since W is the area enclosed by the cycle it follows that $W > 0$.

Cont. . .

- Since $W = Q_1 + Q_2$ and Q_2 is negative it follows that only part of the heat absorbed by the gas at the source T_1 (higher temperature) can be transformed into work.
- The other part is surrendered to the source at T_2 (lower temperature).

Cont. . .

- We define the efficiency of the Carnot cycle, η, as the ratio between the work done and the heat absorbed by the gas at the source T_1.

$$\eta = \frac{Q_1 + Q_2}{Q_1} = 1 + \frac{Q_2}{Q_1} \; (Q_1 > 0, Q_2 < 0)$$

- By considering the relations derived for Q1 and Q2 we can write the above ratio as

$$\eta = 1 + \frac{-nR^* T_2 \ln \frac{V_3}{V_4}}{nR^* T_1 \ln \frac{V_2}{V_1}}$$

$$\eta = 1 - \frac{T_2}{T_1} \dots \dots \dots \dots (3)$$

Cont. . .

- The derivation shows that the efficiency depends **only on the temperatures** of the two heat reservoirs.
- **Volume terms cancel out** because the expansion and compression ratios are the same.
- This result confirms that the **Carnot efficiency is maximized when T2 is as small as possible**

Lessons learned from the Carnot cycle

- According to equation (3) the thermodynamic efficiency of the cycle depends only on the temperatures of the two heat sources and it becomes zero if $T_2 = T_1$ (i.e. if the two sources collapse into one source).
- As such we conclude that doing work via a thermal engine operating at only one source of heat is impossible.
  - ✓ This is known as **Kelvin's postulate** and is an expression of the second law of thermodynamics.

Cont. . .

- ❖ Another way of stating this suggest is that it is impossible to construct an engine which transforms heat into work without surrendering some heat to a source at a lower temperature.
- ❖ According to the second law, apart from the warm heat source (sea or air), a second colder heat source would be required.

Cont. . .

- For example, if the atmosphere, in the horizontal, is thought of as a Carnot cycle operating at $T_1 =$ equatorial temperatures and $T_2 =$ polar temperatures
  - ✓We can explain why in the winter the circulation is stronger. Recall from above that

$$\eta = 1 + \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$$

Cont. . .

- If we think of summer as the limiting case where $T1 - T2 \rightarrow 0$, then during summer $Q2 \rightarrow -Q1$ and $\eta \rightarrow 0$.

- On the other hand if we think of winter as the limiting case where $T2 \ll T1$, then during winter $\eta \rightarrow 1$ and $Q2 \rightarrow 0$.

  - ✓ It follows that the efficiency of this "machine" is higher in the winter than in the summer.

- Accordingly, the amount of heat given by the atmosphere to the cold source is smaller in the winter, than in the summer.

- Since the amount of heat absorbed by the atmosphere from the warm source is more or less constant, this means that there is more heat "remaining" in the atmosphere in the winter than in the summer that can be transformed into kinetic energy.

The Carnot Cycle in Meteorology

- The Carnot cycle is an idealized thermodynamic cycle that represents the most efficient way to convert heat into work. It consists of four stages:
- *Isothermal Expansion:* Heat is absorbed from a warm source at a constant temperature, causing expansion.
- *Adiabatic Expansion:* The system expands without heat exchange, causing cooling.
- *Isothermal Compression:* The system releases heat at a cooler temperature.
- *Adiabatic Compression:* The system compresses without heat exchange, increasing its temperature

Relevance to Meteorology

- The Carnot cycle helps in understanding the efficiency of heat engines in the atmosphere, such as hurricanes and cyclones.
- A hurricane functions like a Carnot engine, where the warm ocean supplies heat, the storm system does work, and energy is lost at the upper atmosphere.
- The difference in temperature between the warm ocean and the cooler upper atmosphere determines the storm's efficiency

Heat Engines

- A heat engine is a device that does work through the agency of heat

Carnot Efficiency and Its Connection to Meteorology

- In meteorology, atmospheric processes often involve energy transfers, which can be analyzed using thermodynamic principles similar to those in Carnot cycles. The equation:
- $\eta = 1 - \frac{T_2}{T_1}$

describes the efficiency of a heat engine, and in meteorology, the atmosphere behaves like a natural heat engine that drives large-scale circulation patterns

Atmospheric Heat Engines: The Carnot Cycle in Meteorology

- The Earth's atmosphere and oceans operate like a giant heat engine, powered by solar radiation. The efficiency of this heat engine can be estimated using the Carnot efficiency equation.
- T1(High Temperature Reservoir): The warm surface of the Earth, heated by solar radiation (e.g., tropical oceans at ~300 K).
- T2 (Low Temperature Reservoir): The colder upper atmosphere or polar regions (e.g., tropopause ~200 K).
- Energy Source: The Sun provides heat, which is absorbed at the surface and later released to space through convection and radiation

Cont. . .

- Using:

$$\eta = 1 - \frac{T_2}{T_1}$$

- For an idealized system with T1=300K (surface) and T2=200K (upper troposphere):

η = 33%Thus, only about one-third of the absorbed energy is converted into mechanical motion (e.g., winds, cyclones), while the rest is lost as heat

Hurricanes as Heat Engines

- Hurricanes (tropical cyclones) behave like **Carnot engines**:
- **T1**: Warm ocean surface (~300 K) acts as a heat reservoir.
- **T2**: Cold upper troposphere (~200 K) acts as the heat sink.
- The efficiency formula helps estimate **how much energy from the ocean is converted into wind speed**.
- A greater temperature difference (higher η results in **stronger storms)**.

Climate Change and Atmospheric Efficiency

- As global warming increases surface temperatures, T1 **rises**, while T2 **may remain constant or decrease** due to enhanced cooling in the upper atmosphere.
- This increases the **efficiency (η) of atmospheric circulation**, potentially leading to **stronger storms and altered wind patterns**

Cont. . .

- The Carnot efficiency formula, originally from thermodynamics, is directly applicable to atmospheric science:
- It explains wind formation, storm intensification, and climate dynamics.
- It shows how energy from the Sun is converted into atmospheric motion.
- It helps predict how climate change might affect atmospheric circulation

The Atmospheric Heat Engines

- ❖The atmosphere itself can be regarded as the working substance of an enormous heat engine.
  - ✓Heat is added in the tropics, where the temperature is high.
  - ✓Heat is transported by atmospheric motions from the tropics to the temperate latitudes.
  - ✓Heat is emitted in temperate latitudes, where the temperature is relatively low.
- ❖We can apply the principles of thermodynamic engines to the atmosphere and discuss concepts such as its efficiency.

….END

Thanks !

Chapter 3: The Second law of Thermodynamics

Objective

- *To Explain the* Entropy in Atmospheric Science
- *To Know Special Forms Of The Second Law*
- *To See ways to combining the first and second laws of Thermodynamics*
- *To Know Reversible and Irreversible Processes in Dry and Moist Air*

Entropy

- A measure of the level of disorder or **randomness** in a system
- entropy, represented by S
- $$dS = \frac{dQ}{T}$$

Where:

dS = change in entropy

δQ= heat added

T = absolute temperature in K

Cont. . .

- Entropy is a measure of disorder, and when the Sun rises, it increases the energy in the atmosphere, causing more mixing, motion, and weather activity leading to greater atmospheric entropy.

Entropy in Atmospheric Science

- According to the Second Law of Thermodynamics, the total entropy of an isolated system tends to increase over time
- This means the atmosphere naturally moves toward a state of greater disorder, affecting weather and climate

Entropy in Meteorology

- In meteorology, entropy helps describe how the atmosphere distributes energy, especially during processes like

❖ Adiabatic Processes

- When air rises or sinks in the atmosphere, it expands or compresses without exchanging heat.
- If the process is reversible and adiabatic, it is isentropic — meaning entropy remains constant.
- The concept of isentropic surfaces is used to track air parcels and understand motion in the upper atmosphere

Cont. . .

- **Air Parcel Stability**
  - Entropy is used to determine atmospheric stability
  - A parcel of air that is warmer and/or more moist (often associated with higher entropy) than its surroundings will be less dense and tend to rise, contributing to convection, cloud formation, and potentially storms.

Cont. . .

- **Weather System Development**
  - Entropy increases during irreversible processes like:
    - Turbulence
    - Friction
    - Mixing of air masses
    - Condensation of water vapor
  - These processes drive the evolution of weather systems, like cyclones and fronts

Cont. . .

- The First Law of Thermodynamics for a reversible transformation may be written as

$$dq = c_p dT - \alpha dp$$

- Therefore,

$$ds = \frac{dq}{T} = c_p \frac{dT}{T} - \frac{\alpha}{T} dp = \left(c_p \frac{dT}{T} - R \frac{dp}{p} \right)$$

- In this form the First Law contains functions of state only.

Exercise

- In a thermodynamic process a parcel of dry air is lifted in the atmosphere such that its pressure decreases from 1000 to 800 mb while its temperature remains constant. Calculate the change in the specific entropy of the parcel.

Special Forms Of The Second Law

- Given the above definitions of the second law we can derive its expression for special cases such as the following

- ***Finite Isothermal Transformation ($\triangle U = 0$)***

$$\triangle S \geq \frac{1}{T}\int_i^f dQ$$

$$\triangle S \geq \frac{Q}{T}$$

$$\triangle S \geq \frac{W}{T}$$

- ***Adiabatic Transformation***

$$ds \geq 0$$

Cont. . .

- ***Iseontropic Transformation***

An isontropic transformation is one during which the entropy is does not change in this case it is clear that

$$dQ \leq 0$$

- ***Isochoric Transformations***

From the first law we have that when dv=0 then dq=Cvdt

so $\quad dS \geq \text{Cv} \ln \frac{dT}{T} \qquad \triangle S \geq \text{Cv} \frac{Tf}{Ti}$

Combining the first and second laws

- Consider the following form of the first law

$$\delta Q = Cpdt - Vdp$$

Combining the above equation with $dS \geq \delta Q/T$ yields

$$TdS \geq CpdT - V\,dp$$

Recalling that $Cp = dH/dT$ we can reduce the above equation to

$TdS \geq dH - V\,dp$ or $dH \leq TdS + V\,dp$

Similarly, from

$$\delta Q = C_V dT + pdV$$

we arrive at

$$TdS \geq dU + \delta W$$

or

$$dU \leq TdS - pdV.$$

Reversible and Irreversible Processes in Dry and Moist Air

- In meteorology, reversible and irreversible processes describe how atmospheric systems evolve under different conditions.
- These processes affect energy exchange, cloud formation, and precipitation

Reversible Process

- A process that can return to its initial state without loss of energy and without increasing entropy in an ideal case.
- Requires no friction, turbulence, mixing, or phase changes with latent heat dissipation.
- Example: Idealized adiabatic ascent/descent of dry air

Irreversible Process

- A process that cannot return to its initial state without external energy input and leads to an increase in entropy.
- Caused by friction, turbulence, mixing, phase changes (condensation, evaporation), and radiative processes.
- Example: Cloud formation, precipitation, turbulent mixing, and radiative cooling

Why Does This Matter

✓ ***Weather Prediction:*** Understanding irreversibility helps meteorologists predict cloud formation, storm development, and turbulence.

✓ ***Climate Science:*** The irreversible processes in the hydrological cycle (precipitation, evaporation) impact global energy balance.

✓ ***Energy Budget:*** The increase in entropy through these processes influences heat transport in the atmosphere

Generalized Statement of 2nd Law

- The Second Law of Thermodynamics states that
- For a *reversible transformation there is no change in the entropy of the universe.*
  - In other words, if a system receives heat reversibly, the increase in its entropy is exactly equal in magnitude to the decrease in the entropy of its surroundings.

Cont. . .

- ❖One way of stating the Second Law of Thermodynamics is as follows: Heat can be converted into work in a cyclic process only by transferring heat from a warmer to a colder body.
- ❖Another statement of the Second Law is: Heat cannot of itself pass from a colder to a warmer body in a cyclic process.
- ❖That is, the “uphill” heat-flow cannot happen without the performance of work by some external agency.

Cont. . .

- ❖The concept of reversibility is an abstraction.
- ❖A reversible transformation moves a system through a series of equilibrium states so that the direction of the transformation can be reversed at any point by making an infinitesimal change in the surroundings.
- ❖All natural transformations are irreversible to some extent.

Cont. . .

- In an irreversible (sometimes called a spontaneous) transformation, a system undergoes finite transformations at finite rates, and these transformations cannot be reversed simply by changing the surroundings of the system by infinitesimal amounts.
  - Examples of irreversible transformations are the flow of heat from a warmer to a colder body, and the mixing of two gases.

Cont. . .

- If a system receives heat dq_{irrev} at temperature T during an irreversible transformation, the change in the entropy of the system is not equal to $\frac{dq_{irrev}}{T}$.
- In fact, for an irreversible transformation there is no simple relationship between the change in the entropy of the system and the change in the entropy of its surroundings.
- However, the remaining part of the second law of thermodynamics states that the *entropy of the universe increases as a result of irreversible transformations.*

Cont. . .

❖ The two parts of the second law of thermodynamics stated earlier can be summarized as follows

$$\Delta s_{universe} = \Delta s_{system} + \Delta s_{surroundings}$$

$$\Delta s_{universe} = 0 \; for \; reversible \; (equilibrium) transformations$$

$$s_{universe} > 0,$$

for irreversible (spontaneous)transformation

Why Natural Processes Are Irreversible?

- Even though natural meteorological processes are mostly irreversible, reversible processes are still important because they provide a theoretical framework to
  - Simplify Complex Atmospheric Processes
  - Define Reference States for Understanding Deviations
  - Help in Thermodynamic Efficiency and Energy Budget Studies
  - Improve Numerical Weather Prediction (NWP) Models

Cont. . .

- We study reversible processes to build a foundation for understanding real, irreversible meteorological phenomena.

  ✓ Reversible models simplify reality, allowing us to make predictions before accounting for complexity.

  ✓ Even though real weather is messy and irreversible, the insights from reversible process theory are essential for weather forecasting, climate science, and atmospheric physics

….END

Thanks !